

PROJECT REPORT

Economic Survey Management System Using C Language

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Introduction

The “Economic Survey Management System” is a C language project developed to manage and analyze household economic data. This system stores information about different households including their income, expenses, and savings.

The project helps in performing simple economic analysis such as:

- Average income calculation
- Highest income identification
- Poverty line checking
- Household searching

This project demonstrates the practical implementation of programming concepts along with basic economic data analysis.

Objectives

- To store household economic records
- To calculate savings
- To analyze household income
- To search records efficiently
- To practice C language concepts

Features of the Project

1. Add household data
2. Display all household records
3. Calculate average income
4. Find highest income household
5. Search household by ID
6. Poverty line analysis

Concepts Used

Concept	Description
Structures	Used to store household data

Concept	Description
Arrays	Used for storing multiple records
Functions	Used for modular programming
Loops	Used for repetition
Conditional Statements	Used for decision making
Menu Driven Program	Used for user interaction
File handling	Used to store data

Structure Used

```
struct household
{
    int id;
    float income;
    float expense;
    float savings;
};
```

Functions Used

Function Name	Purpose
inputData()	Takes household data
displayData()	Displays all records
averageIncome()	Calculates average income
highestIncome()	Finds highest income
searchHouse()	Searches household by ID
saveToFile()	To save file
loadFromFile()	To load file
isDuplicate	Used not to take duplicate id

Working of the Project

1. User enters number of households.
2. User enters household data:
 - ID
 - Income
 - Expense
3. Savings are automatically calculated.
4. Menu options allow different operations.
5. User can search records and analyze data.

Savings Formula

Savings are calculated by:

$$\text{Savings} = \text{Income} - \text{Expense}$$

Poverty Line Checking

If household income is less than 30000, the system displays:

Below Poverty Line

Otherwise:

Economically Stable

Sample Output

```
=====
ECONOMIC SURVEY MANAGEMENT SYSTEM
=====
```

Enter number of households: 2

Enter data for household 1

Enter household ID: 101

Enter monthly income: 50000

Enter monthly expense: 30000

Enter data for household 2

Enter household ID: 102

Enter monthly income: 25000

Enter monthly expense: 20000

Advantages of the Project

- Simple and user friendly
- Easy economic data analysis
- Demonstrates major C concepts
- Helpful for beginners
- Useful for academic learning

Limitations

- Limited statistical analysis
- No graphical interface

Conclusion

The Economic Survey Management System is a useful C language project for managing and analyzing household economic information. The project combines programming concepts with economic data analysis and provides practical understanding of structures, arrays, functions, and loops in C language.

It is suitable for beginners and academic project submissions.

Tools and Technologies

- C Language
- Dev C++